SDE Roadmap

Stage	Primary Focus	New Skills to Learn
Student → Intern	Learn to code	Programming language, DSA, Git, OOP, SQL, basic web/app development, debugging, Linux basics
Intern → Associate Software Engineer	Build production code	Frameworks (React/Spring/Node/.NET), REST APIs, databases, testing, code reviews, Docker basics, cloud basics, Agile, communication
Associate SE → SDE 1	Own features independently	System design basics, API design, authentication, caching, CI/CD, logging, monitoring, performance optimization, writing documentation, estimating tasks
SDE 1 → SDE 2	Design systems	Distributed systems, message queues, microservices, concurrency, scalability, security, cloud deployment, Kubernetes basics, mentoring juniors
SDE 2 → Senior SDE	Lead projects	Advanced system design, architecture patterns, database optimization, incident handling, technical leadership, stakeholder communication, project planning
Senior SDE → Staff Engineer	Influence multiple teams	Cross-team architecture, design reviews, organization-wide standards, performance tuning at scale, technical strategy, mentoring senior engineers
Staff → Principal	Company-wide impact	Long-term architecture, technology strategy, platform engineering, cost optimization, innovation, executive communication, technical vision
Principal → Distinguished/Fellow	Industry leadership	Research, patents, open source leadership, speaking at conferences, defining engineering direction, mentoring principals

Detailed Skill Checklist

1. Student → Intern

Programming

- Python, Java, C++, or JavaScript
- Variables, loops, functions
- Object-Oriented Programming
- Exception handling

Computer Science

- Arrays, Linked Lists
- Stacks & Queues
- Trees
- Graphs
- Hash Maps
- Sorting & Searching
- Recursion
- Dynamic Programming (basic)

Development

- Git & GitHub
- SQL
- HTML/CSS/JavaScript (or mobile development)
- Linux commands
- VS Code/IDE usage
- Debugging

Projects

- 3–5 portfolio projects
- GitHub profile
- Resume

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2. Intern → Associate Software Engineer

Backend

- REST APIs
- CRUD applications
- Authentication (JWT/OAuth)
- File uploads
- Error handling

Database

- MySQL/PostgreSQL
- MongoDB
- Indexing
- Transactions

Tools

- Docker
- Postman
- Git branching
- Unit testing
- API testing

Cloud

- AWS/GCP/Azure basics
- Deploy applications

Professional Skills

- Read existing code
- Participate in code reviews
- Write clean code
- Write documentation

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3. Associate SE → SDE 1

Now companies expect **independence**.

Learn:

- System Design fundamentals
- Database design
- API versioning
- Redis caching
- Background jobs
- Message queues
- CI/CD
- Logging
- Monitoring
- Security basics
- Design patterns
- SOLID principles
- Performance optimization
- Code refactoring

Also:

- Break features into tasks
- Estimate timelines
- Review pull requests
- Write technical documentation

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4. SDE 1 → SDE 2

Focus on **scalability**.

Learn:

- Microservices
- Kubernetes
- Distributed systems
- CAP theorem
- Load balancing
- Reverse proxy
- CDN
- WebSockets
- Event-driven architecture
- Kafka/RabbitMQ
- Multithreading
- Concurrency
- Database replication
- Sharding
- Observability
- Security best practices

Responsibilities:

- Design complete modules
- Mentor interns
- Improve team productivity

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5. SDE 2 → Senior SDE

Now you're expected to **lead technical execution**.

Skills:

- Large-scale system design
- Architecture decisions

- Technical leadership
- Production debugging
- Incident response
- Performance optimization
- Cost optimization
- Design reviews
- Cross-team collaboration
- Mentoring

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6. Senior → Staff

Responsibilities shift from building systems to shaping engineering direction.

Learn:

- Platform engineering
- Engineering standards
- Organization-wide architecture
- Cloud cost optimization
- Developer experience
- Long-term planning
- Cross-team influence
- Technical strategy

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7. Staff → Principal

Focus:

- Company-wide architecture
- Technical vision
- Build platforms used by many teams
- Technology evaluation
- Engineering strategy
- Innovation
- Executive communication

Skills That Matter at Every Stage

- Communication
- Problem-solving

- Debugging
- Git proficiency
- Clean code
- Reading documentation
- Writing documentation
- Time management
- Collaboration
- Learning new technologies quickly

If your goal is to work at top product companies such as Google, Microsoft, Amazon, or Meta, a strong progression is:

1. Master DSA and one programming language.
2. Build full-stack projects.
3. Learn backend development and databases.
4. Learn cloud, Docker, and CI/CD.
5. Learn system design.
6. Learn distributed systems.
7. Gain leadership through code reviews, mentoring, and architecture.

This sequence aligns well with how engineers typically progress from intern to senior technical roles.